

Titanic Survival Prediction Data Science Report

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1.Problem Description

This project focuses on analyzing the Titanic dataset and building a machine learning model to predict whether a passenger survived or not. The primary objective is to understand data analysis, visualization, and machine learning by applying them to a real-world dataset. It covers the complete data science pipeline from data preprocessing to visualization and prediction.

2.Learning Outcomes

Through the completion of this project, several key data science skills were applied and demonstrated:

1. Data Cleaning and Preprocessing using NumPy and Pandas.
2. Data Visualization using Seaborn and Matplotlib.
3. Feature Engineering.
4. Building Machine Learning Models.
5. Model Evaluation using Accuracy and Confusion Matrix.

3. Data Processing Pipeline

1. Data Collection

The Titanic dataset was loaded into a Pandas DataFrame for analysis.

2. Data Cleaning

Missing values were handled appropriately: Missing values in the age column were filled using the mean age while missing embarked values were filled using the mode.

Unnecessary columns were removed from the dataset to streamline the model. The dropped columns included class, who, adult_male, deck, embark_town, and alive.

Categorical variables like sex, embarked, and alone were mapped to numerical values to prepare them for the machine learning algorithm.

3. Exploratory Data Analysis and Visualization

Various graphs like bar plots, histograms, and heatmaps were created to understand relationships within the data.

Visualizations generated included survival rates by gender and family size, fare distribution by passenger class, and overall age distribution comparisons.

4. Feature Engineering

New features were created to extract better insights, such as FamilySize. This was calculated by combining the number of siblings/spouses (sibsp) and parents/children (parch).

Key features retained for model training included Age, Sex, Pclass, Fare, and FamilySize.

4. Model Building and Evaluation

1. Machine Learning Model

Logistic Regression was used to classify whether a passenger survived or not based on input features.

The data was split into training and testing sets (80% training, 20% testing) before fitting the model.

2. Results

The Logistic Regression model achieved an accuracy of around 80.49%.

The confusion matrix showed that most predictions were correct.

Model evaluation relied heavily on standard metrics like Accuracy and the Confusion Matrix to validate the classifications.

5. Conclusion

This project successfully demonstrates how data science techniques can be used to extract insights and make predictions. By executing tasks ranging from handling missing data and engineering new features to deploying a Logistic Regression classifier, this project provides a complete understanding of the data science workflow.